



Does utilizing WHO's interim targets further reduce the risk - meta-analysis on ambient particulate matter pollution and mortality of cardiovascular diseases?☆

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ABSTRACT

Long-term exposed to ambient particulate matter (PM) has been recognized as a risk factor for cardiovascular diseases (CVDs) mortality but few studies examine the utility of WHO's interim targets (2006) concerning PM control on CVD mortality. This review aimed to synthesize the long-term exposed to ambient PM exposures on overall CVD mortality according to WHO's interim targets; meanwhile, subgroup analysis by ethnicity, smoking status and PM assessment method were also conducted. We systematically searched studies published between January 1974 and Jul 2017 in PubMed and Embase. Quality of each study was assessed using Critical Appraisal Skill Programme (CASP) checklist. Fixed-effects or random-effects model of meta-analysis was determined by the test of heterogeneity. Subgroup analyses were conducted according to ethnicity, smoking status, PM assessment method and interim PM targets of WHO guidelines. Overall, 16 eligible studies were included, covering 15,511,997 participants and 542,991 CVD deaths. Five studies concerning PM_{2.5} were rated as good quality. The pooled hazard ratio (HR) of every 10 μ g/m³ increment of PM_{2.5} exposure for CVD mortality was 1.12 (95% CI = 1.08–1.16), but it was not significant for PM₁₀ (HR = 1.02, 95%CI = 0.89–1.16). Compared with the annual PM_{2.5} exposure level within WHO's interim targets (1.11–1.16), significantly smaller HR was observed for subjects with an exposure level below WHO's air quality guideline (HR = 1.03, 95%CI = 1.02–1.04). The pooled HR was relatively higher for studies in Asian and with at least 11 years' follow-up and those adopting relatively poor methods (category 1) in assessing PM_{2.5}, whilst the risk was similar regardless of smoking status. Egger and Begg's tests showed no evidence for publication bias. Long-term ambient PM_{2.5} exposure level was positively associated with the overall CVD mortality. Different interim targets above the WHO's Air Quality Guideline level exerted a similar scale of CVD risk, but there is no evidence for a threshold.

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1. Introduction

Cardiovascular diseases (CVDs) are the leading causes of death that accounted for nearly 31% of global mortality in 2012 (World Health Organization, 2006). Particulate matter (PM) is a complex mixture with a number of components including acids, organic chemicals, and metals (Hildemann et al., 1991). Many of these components are highly toxic substances associated with chronic diseases including CVDs (Sacks et al., 2011). According to the

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research done by Schauer, the source of PM mainly came from diesel and gasoline exhaust, emission from cooking and heating, dust, cigarette smoke and natural gas combustion (Schauer et al., 1996). PM could be divided into PM_{2.5} (aerodynamic diameter <2.5 µm) and PM₁₀ (aerodynamic diameter ≤10 µm), and the harmful effect of PM on CVDs was firstly reported in the Harvard 6 cities' study, in which an increased risk of cardiopulmonary disease mortality was observed among residents with long-term PM exposure ranged from 11.0 to 29.6 µg/m³ (Dockery et al., 1993; Grahame and Schlesinger, 2010). This finding was further supported by several studies that were conducted in France, Netherlands, and UK (Bentayeb et al., 2015; Brunekreef et al., 2009; Carey et al., 2013). Long-term exposure to elevated level of PM has posed serious public health problems worldwide, particularly in the areas with heavy PM pollution (Villeneuve et al., 2015).

The most recent meta-analysis regarding the long-term exposure to ambient PM air pollution and CVD mortality was conducted by Hoek et al., in 2013. They reported a 11% increased risk [95% confidence interval (95%CI) = 6–13%] of CVD mortality for per 10 µg/m³ increase of PM_{2.5} and 3.5% increased risk (95%CI = 0.4%–6.6%) for PM₁₀ (Hoek et al., 2013). However, the review of Hoek et al. combined occupational and non-occupational studies and thus their results are probably entangled by occupational source of exposure to PM_{2.5} (Fang et al., 2010). World Health Organization (WHO) set a series of interim targets for PM guideline (World Health Organization, 2006) based on the accumulative evidence on the harmful effects of PM_{2.5} aiming at reducing disease burden. Although this WHO's interim target for PM guideline quantitatively measures possible gradient relations with PM exposures, no review has been conducted to summarize the effect of PM on CVD mortality according to different interim targets of WHO. Also, there are possible variations in the effect of PM_{2.5} on CVD mortality amongst subgroups of ethnicity, smoking status, exposure period and PM assessing method. As vulnerable populations, such as, smokers or people with longer time PM exposure are more susceptible for developing cardiovascular diseases and premature death; however, these areas of evidence have never been assessed and summarized.

This review aimed to synthesize the long-term effect of ambient PM air pollution on overall CVD mortality according to WHO's interim targets (World Health Organization, 2006); meanwhile, subgroup analysis by ethnicity, smoking status and PM assessment method were also conducted.

2. Methods

This review follows the guidelines of the PRISMA statement (Moher et al., 2009).

2.1. Eligibility and search strategy

We searched PubMed and Embase databases for cohort studies that reported the association between CVD mortality and long-term exposure (study period equal or larger than 1 year) to outdoor PM₁₀ or PM_{2.5}. The search included papers published in English and between January 1974 and July 2017. Detailed searching strategy was summarized in Supplement 1. Two authors (ZL and FW) independently reviewed potential studies for inclusion, and any discrepancy encountered was resolved by consensus.

2.2. Inclusion and exclusion criteria

Duplicated studies were excluded. The initial screening was based on the title and abstract of searching results. The following inclusion criteria were used for the screening: (i) investigated the association between CVD mortality and long-term ambient PM

exposure; (ii) individual studies provided sufficient exposure data on PM₁₀ or PM_{2.5}; and (iii) cohort study; and (iv) provided sufficient data to calculate hazards ratio (HR) with 95%CI. Studies that met the following items were excluded: (i) irrelevant to PM or CVD mortality; (ii) study period shorter than 1 year; (iii) indoor rather than outdoor PM exposure; (iv) non-cohort study. Full-text papers were reviewed to determine the eligibility of studies if their abstracts missed the key information. Studies were excluded if they did not report the quantitative estimation of effect size for the association between PM₁₀ or PM_{2.5} and CVD mortality (or incidence). Studies were also excluded if they focused on CVD prognosis among patients or specific occupational groups. If two or more reports were generated from one cohort, the report that presented the largest sample or longest follow-up period was included in this review.

2.3. Data extraction

We extracted information from each eligible study regarding the study population/region, study period, exposure assessment (i.e., mean concentration and concentration range; data source and exposure assessment method), definition of outcome (ICD codes, 390–459 for ICD-9, or I00–I99 for ICD-10), number of CVD deaths, and hazard ratio (HR) (after adjustment) for each 10 µg/m³ increment of exposure level and its 95% CI. We re-calculated the weighted arithmetic mean concentration for the Nishiwaki's study according to the population size and mean concentration of PM in each city due to lack of relevant exposure data (Nishiwaki et al., 2013). Two reviewers (ZL and YW) extracted the data independently.

2.4. Quality assessment

Quality assessment was conducted for each study using Critical Appraisal Skills Programme (CASP) checklist (Claydon, 2015). The checklist for assessing study quality provides an overall score for each study and subscales for three individual sections: study valid (8 items), result condition (3 items) and result applicability (3 items). Items regarding the cohort assessment were adopted for quality assessment, yielding a maximal score of 14 for each study based on the checklist (Supplement 2). We classified an included study as relatively low quality if the score of this study was below or equal to the median score; otherwise, the study was rated to have good quality. In CASP checklist, the Question 3 (Q3) "Was the exposure accurately measured to minimize bias?" was to evaluate the quality of exposure assessment methods (Supplement 2 & 3). According to PM assessment method described in the methodology studies (Hoek, 2017), we assigned "1" for the individual studies that retrieved PM exposure data in address level based on chemical transportation mode, land use regression modeling, or satellite modeling (category 3). We assigned "0.5" for those retrieving PM exposure data using interpolation concentration, atmospheric diffusion, or dispersion model in community level based on multiple fixed monitoring stations (category 2). We assigned "0" for the remaining studies that retrieved PM data at city level based on a single fixed monitoring station (category 1). Two authors (ZL and YW) independently scored each included study and any discrepancy encountered was resolved by consensus.

2.5. Statistical analysis

We conducted meta-analysis by using Stata v14.1 (Stata Corp, College station, Texas, USA) to obtain the pooled HRs for per 10 µg/m³ increment of PM₁₀ and PM_{2.5}, respectively (Deeks et al., 2001). Random-effects model was used to combine HRs from individual

studies if significant heterogeneity existed amongst studies (i.e., Cochran's Q test, p -value < 0.10). Fixed-effects model was used if the test for heterogeneity was not statistically significant (DerSimonian and Laird, 1986; Dickersin and Berlin, 1992). Sub-group analyses were performed to examine whether the risk of exposure to ambient PM_{2.5} and PM₁₀ varied by different annual exposure levels according to the WHO's air quality guideline for different Interim targets [Interim target (IT) 1 to 3, (IT1: PM_{2.5} < 35 $\mu\text{g}/\text{m}^3$ and PM₁₀ < 70 $\mu\text{g}/\text{m}^3$; IT2: PM_{2.5} < 25 $\mu\text{g}/\text{m}^3$ and PM₁₀ < 50 $\mu\text{g}/\text{m}^3$; IT3: PM_{2.5} < 15 $\mu\text{g}/\text{m}^3$ and PM₁₀ < 30 $\mu\text{g}/\text{m}^3$ or air quality guideline (AQG) PM_{2.5} < 10 $\mu\text{g}/\text{m}^3$ and PM₁₀ < 20 $\mu\text{g}/\text{m}^3$] (World Health Organization, 2006). We further examined whether the effect of PM₁₀ or PM_{2.5} varied amongst subgroups of cigarette smoking, geographic regions, exposure assessment methods of PM, follow-up years, and study quality. Sensitivity analysis was performed to test the robustness of results by excluding the study with the largest sample size. Possible publication bias was tested by Egger and Begg's tests (JPT CHH, 2011).

3. Results

3.1. Literature search and characteristics of studies

In total, 16 studies were judged as eligible and included in the meta-analysis (Fig. 1). Characteristics of the included studies are shown in Table 1 (Beelen et al., 2014; Brunekreef et al., 2009; Carey et al., 2013; Cesaroni et al., 2013a; Crouse et al., 2015; Fischer et al., 2015; Hales et al., 2010; Lepeule et al., 2012; Miller et al., 2007; Nishiwaki et al., 2013; Pope et al., 2015; Thurston et al., 2016; Ueda et al., 2012; Villeneuve et al., 2015; Wong et al., 2015; LW Zhang et al., 2014). However, Beelen's publication is a systematic review in which the PM exposure data were reconstructed by Beelen et al., as no PM exposure data was provided by the original studies that

were included in Benlen's review.

A total of 542,991 CVD deaths were observed among 15,511,997 participants. Two studies separately reported the effect of PM_{2.5} and PM₁₀ on CVD mortality (Carey et al., 2013); 10 studies only reported the effect of PM_{2.5}, while 4 studies reported the association with PM₁₀. All studies included both sexes except for the Miller's study, which only included postmenopausal women (Miller et al., 2007). Among 11 studies concerning ambient PM_{2.5}, 7 of them were performed in North America, 3 in Europe, and 1 in Asia. Among 5 studies regarding PM₁₀, 1 was conducted in Europe, 1 in Oceania, and 3 in Asia. The duration of follow-up period ranged from 4 to 36 years with a median period of 11 years for PM_{2.5} and 12 years for PM₁₀. The annual exposure concentration varied in-between 8.9–35.3 $\mu\text{g}/\text{m}^3$ for PM_{2.5} and 8.4–144 $\mu\text{g}/\text{m}^3$ for PM₁₀.

3.2. Assessment methods of PM exposure and CVD mortality

PM exposure assessment methods were classified by monitoring techniques and exposure measurement status (Dong and Wu, 2017; Hoek, 2017). Briefly, category 1 was classified based on one fixed monitoring station to represent population exposure level; category 2 was classified as these studies provided geographic information such as applying interpolation method from multiply monitoring station or atmospheric diffusion or dispersion model; category 3 contains relatively high-accuracy models which provided more accurate community or household based estimation on PM concentration, such as, land use regression model, chemical transformation model or other satellite model. Five studies listed the spatial area of grid cell with a resolution value from 1*1 km² to 10*10 km² (Carey et al., 2013; Cesaroni et al., 2013a; Crouse et al., 2015; Villeneuve et al., 2015; Wong et al., 2015). Only the Lepeule study used central monitoring data to represent the whole area.

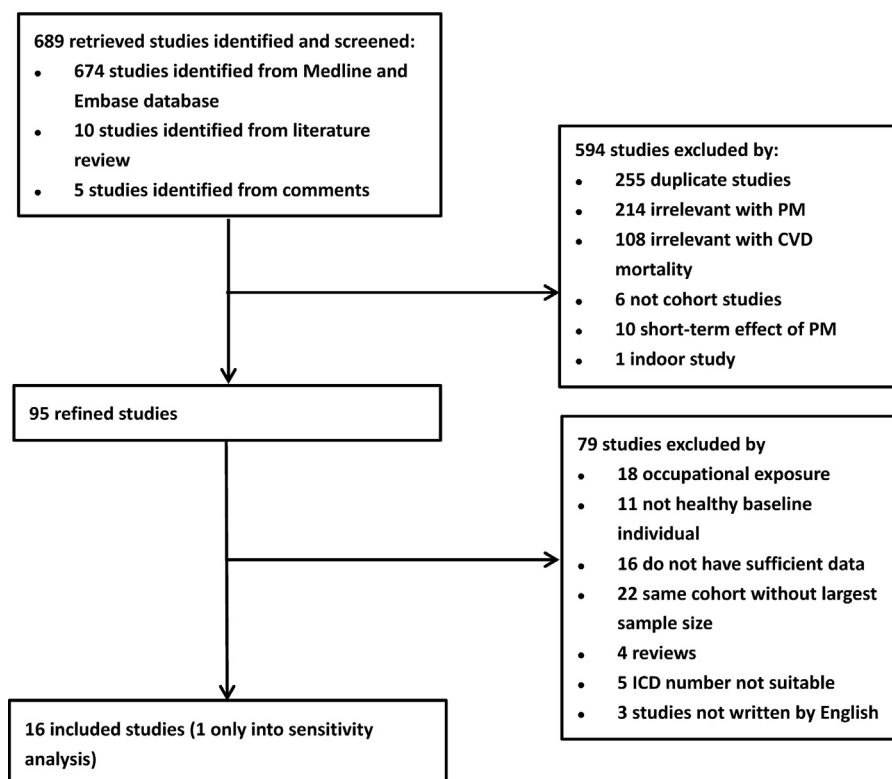


Fig. 1. Selection process of articles in meta-analysis.

Table 1

Characteristics of 16 cohort studies regarding association between long-term ambient PM exposures and risk of all-cause CVDs mortality.

Author (Year)	Baseline age	Region (Follow-up period, years)	CVD cases	Total participants ^a	PM level ($\mu\text{g}/\text{m}^3$) Mean (range)	PM Assessment Methods(spatial scale) ^c	CVDs definition (ICD9/ICD10)	Hazards ratio (per $10\mu\text{g}/\text{m}^3$ increase)	QAS ^d
PM_{2.5}									
Miller 2007	50–79	United States (1994–2002, 9 years)	261	65,893	13.5 (3.4–28.3)	Category 2 (Community level)	N.A.	1.76 (1.25–2.47)	12
Brunekreef 2009	55–69	Netherland (1986–1996, 11 years)	6279	120,852	28.3 (23.0–37.0)	Category 3 (Address level)	400–440/I10–I70	1.04 (0.90–1.21)	13
Lepeule 2012	25–74	United States (1974–2009, 36 years)	3303	8096	15.9 (N.A.)	Category 1 (City level)	400.0–440.9/I10.0–I70.9	1.26 (1.14–1.40)	13
Carey 2013	40–89	United Kingdom (2003–2007, 5 years)	10,583	830,842	12.9 (8.5–20.2)	Category 2 (1-Km grid)	I00–I99	1.23 (1.11–1.36)	12
Cesaroni 2013	≥30	Italy (2001–2010, 10 years)	60,318	1,265,058	23.0 (7.2–32.1)	Category 2 (Address level 1-Km grid)	390–459	1.06 (1.04–1.08)	12.5
Crouse 2015	≥25	Canada (1991–2001, 10 years)	98,970	2,145,400	8.9 (0.9–17.6)	Category 3 (Address level 10-Km grid)	390–459	1.03 (1.02–1.04)	14
Fischer 2015	≥30	Netherland (2004–2011, 8 years)	209,940	7,218,363	29(5 th –95 th : 24–32)	Category 3 (Post-code level)	I00–I99	1.06(1.04–1.08)	13.5
Pope 2015	≥30	United States (1982–2000, 19 years)	100,149	1,184,587	12.6 (1.0–28.0)	Category 3 (Address level)	390–459/I00–I99	1.14 (1.12–1.17)	14
Villeneuve 2015	40–59	Canada (1980–2005, 26 years)	1940	89,835	9.5 (1.3–17.6)	Category 3 (Post-code level 10-Km grid)	400–440/I00–I99	1.32 (1.14–1.52)	12.5
Wong 2015	≥65	Hong Kong (1998–2011, 14 years)	4656	66,820	35.3 (26.4–44.6)	Category 3 (Address level 1 Km-grid)	I00–I99	1.22 (1.08–1.39)	14
Thurston 2016	50–71	United states (2000–2009, 10 years)	26,009	517,041	10.2 (2.9–28.0)	Category 3	I00–I99	1.10 (1.05–1.15)	14
(Beelen 2014) ^e	Not given	European countries(2008–2011, 4 years)	9994	367,383	6.6–31.0	Category 3	I10–I70	1.04(0.93–1.17)	
PM₁₀									
Hales 2010	30–74	New Zealand (1996–1999, 4 years)	7278	1,054,731	8.4 (0–19.0)	Category 2	393–438	1.06 (1.01–1.12)	12.5
Carey 2013	40–89	United Kingdom (2003–2007, 5 years)	10,583	830,842	19.7 (12.6–29.8)	Category 2 (1-Km grid)	I00–I99	1.14 (1.07–1.21)	12
Ueda 2012	≥30	Japan (1985–2004, 20 years)	571	7250	28.0 (N.A.)	Category 2 (Community level)	390–459/I00–I99	0.90 (0.81–1.00)	12
Nishiwaki 2013	40–69	Japan (1990–2005, 16 years)	1464	78,057	25.0 (N.A.) ^b	Category 2 (Community level)	I20–I25, I60–I69	0.78 (0.71–0.87)	12
Zhang 2014	N.A.	China (1998–2009, 12 years)	677	39,054	144.0 (91.0–274.0)	Category 2 (Community level)	I00–I99	1.23 (1.19–1.26)	13.5
(Beelen 2014) ^e	Not given	European countries(2008–2011, 4 years)	9994	367,383	6.6–31.0	Category 3	I10–I70	1.04(0.92–1.16)	

^a The total participant calculated by the number who was exposure to PM_{2.5} or/and PM₁₀.^b The mean concentration was calculated by weighted average exposure from individuals.^c Category 1: fixed monitoring station concentration; category 2: interpolation concentration or atmospheric diffusion or dispersion model; category 3: chemical transportation mode, land use regression model, satellite model.^d Quality assessment score(QAS) is better if the score is high.^e Only considered in sensitivity analysis.

Almost all studies used ICD-9 or ICD-10 to define CVD outcomes except for the study of Miller et al., which was based on medical records and death certificates issued by hospitals (Curb et al., 2003). However, the definition of CVDs varied amongst included studies. Two studies only included a part of circulatory diseases by focusing on I00–I70 of the ICD-10 code (Brunekreef et al., 2009; Lepeule et al., 2012). Nishiwaki et al. only included stroke and coronary heart disease (CHD), and hence it covered a relatively narrower range of CVDs than other studies (Nishiwaki et al., 2013).

As shown in Supplement 4, all included studies adjusted for potential confounding factors of CVDs, including age (13 studies), sex (12 studies) and smoking status (11 studies). However, only two of them further adjusted for systolic blood pressure (SBP) and cholesterol level (Miller et al., 2007; Ueda et al., 2012). Body mass index (BMI), being an important risk factor for CVDs, was also controlled in 10 studies (Carey et al., 2013; Hales et al., 2010; Lepeule et al., 2012; Miller et al., 2007; Nishiwaki et al., 2013; Pope et al., 2015; Thurston et al., 2016; Ueda et al., 2012; Villeneuve et al., 2015; Wong et al., 2015; LW Zhang et al., 2014). Education (11 studies) and income (7 studies) were also considered in most included studies.

3.3. Quality assessment of each individual study

Table 1 summarizes the quality assessment score (QAS) of included studies. Detailed scores of specific aspects are shown in Supplement 3. Scores of included studies ranged from 12 to 14 (85.7%–100% of full score), with a median score of 13 (92.9% of full score). Five studies concerning PM_{2.5} had good quality while only one for PM₁₀ was rated as relatively poor quality. Most included studies considered potential confounding of age and sex; whilst most of the low-quality studies did not describe the rate of lost to follow-up, did not use new air pollution exposure estimation methods at baseline, did not provide the exact P-value for outcome or did not consider important confounding factors.

3.4. Pooled effects of CVD mortality from exposure to ambient PM_{2.5} and PM₁₀

Because we observed significant heterogeneities amongst studies ($I^2 = 94.5\%$, $P < 0.01$ for Q test), we used random-effects model to summarize the pooled effects. Fig. 2 shows the pooled effects of all studies. Exposure to PM_{2.5} significantly increased CVD

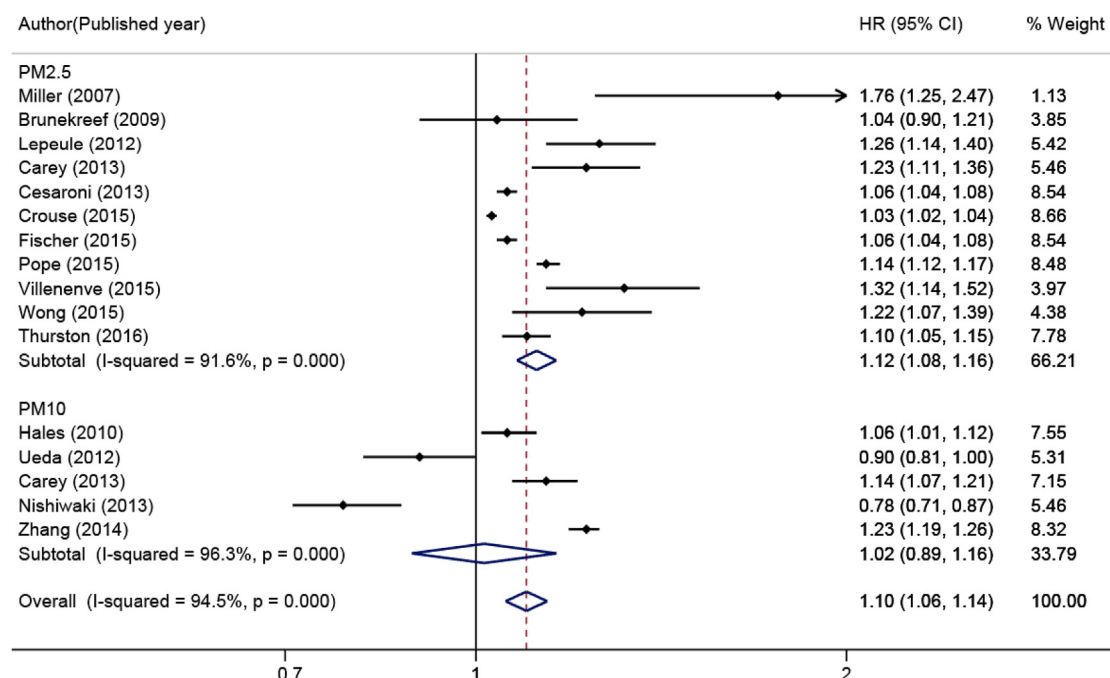


Fig. 2. Meta-analysis: hazard ratios of CVDs mortality for each 10 µg/m³ increment of PM.

mortality by 12% (HR = 1.12, 95%CI = 1.08–1.16, $P < 0.01$ for Q test, $I^2 = 91.6\%$) for every 10 µg/m³ increment. However, there was no significant association between PM10 exposure and CVD mortality (HR = 1.02, 95%CI = 0.89–1.16, $P < 0.01$ for Q test, $I^2 = 96.3\%$). To estimate the effect size of PM with different air pollution levels, we regrouped all studies according to the WHO's air quality guidelines (World Health Organization, 2006). Studies were grouped according to the interim targets (ordinal annual mean concentrations of PM) of WHO (Fig. 3a <IT1, Fig. 3b <IT2, Fig. 3c <IT3, Fig. 3d <AQG). Similar elevated pooled effects of PM2.5 on CVD mortality were observed among subgroups with different WHO's interim targets [Fig. 3(a)–(c)]. Only two studies reported that the risk of PM2.5 under AQG with a statistically insignificant combined HR of 1.15 (95%CI = 0.91–1.47) for each 10 µg/m³ increment. There was lack of a clear relationship between increasing level of ambient PM10 exposure and the hazard ratios of CVD mortality.

3.5. Role of smoking in the effect of PM on CVD mortality

Five studies (3 for PM2.5 and 2 for PM10) provided effect estimates stratified by smoking status (Fig. 4). An increased pooled risk of CVD mortality (per 10 mg/m³ increment of ambient PM2.5) was observed among never smokers (HR = 1.18, 95%CI = 1.06–1.32) and ever smokers (HR = 1.21, 95%CI = 1.06–1.40). However, the pooled effect of PM10 on CVD mortality was not significantly increased among smokers and never smokers.

3.6. Role of geographic region, duration of follow-up, PM assessment methods, and study quality in the effect of PM on CVD mortality

Fig. 5 shows the pooled effects of subgroup analyses by different geographic regions of cohort studies (Fig. 5a), follow-up period (Fig. 5b) and PM2.5 assessment methods (Fig. 5c). There were significant associations between PM2.5 exposure and CVD mortality in North America (HR = 1.17, 95%CI = 1.09–1.25) and Asia (HR = 1.22, 95%CI = 1.07–1.39 in a single study), but a much weaker association

was found in Europe (Fig. 5a). An increased risk of PM10 was observed in a single study in Oceania (HR = 1.06, 95%CI = 1.01–1.12) and Europe (HR = 1.07, 95%CI = 1.07–1.21 in a single study) but not in Asia (HR = 0.96, 95%CI = 0.70–1.30). Two studies in Japan (Nishiwaki et al. and Ueda et al., in Supplement 5) observed a negative effect of PM10 on CVD mortality (HR = 0.84, 95%CI = 0.73–0.96), which contradicted to the reports from China, UK and New Zealand (HR = 1.14, 95%CI = 1.04–1.26).

Fig. 5b shows pooled HRs stratified by follow-up length. According to the median of follow-up, we divided longer term exposure as ≥ 11 years for PM2.5 and ≥ 12 years for PM10. The weighted average annual concentrations of PM2.5 were slightly higher for studies with a longer follow-up period (14.8 µg/m³) than those with a shorter follow-up (13.7 µg/m³), with a corresponding pooled HR of 1.19 (95%CI = 1.11–1.27) and 1.07 (95%CI = 1.04–1.11), respectively.

Effects of PM on CVD mortality stratified by PM2.5 assessment methods are summarized in Fig. 5c. The same category of exposure assessment method was used for PM10. The pooled effect was relatively higher in studies with category 1 (HR = 1.26, 95%CI = 1.14–1.40) and category 2 (HR = 1.22, 95%CI = 1.02–1.46) than those using better exposure assessment methods (For category 3, HR = 1.10, 95%CI = 1.05–1.15).

3.7. Sensitivity analysis

All studies were divided into higher and lower quality subgroups according to the median score value of quality assessment (CASP score ≤ 13 as lower score for PM2.5; CASP score < 12.5 for PM10 exposure). Despite both subgroups showed a significant association between PM2.5 exposure and CVD mortality (Supplement 6), the HR of higher quality studies was lower than that of the low quality studies. However, the pooled effects of both subgroups were not significant for PM10 studies. We calculated pooled effects by excluding the study with the largest sample size to detect if results was significantly distorted by specific studies (Supplement 7), and the results remained stable. We incorporated the pooled HR of

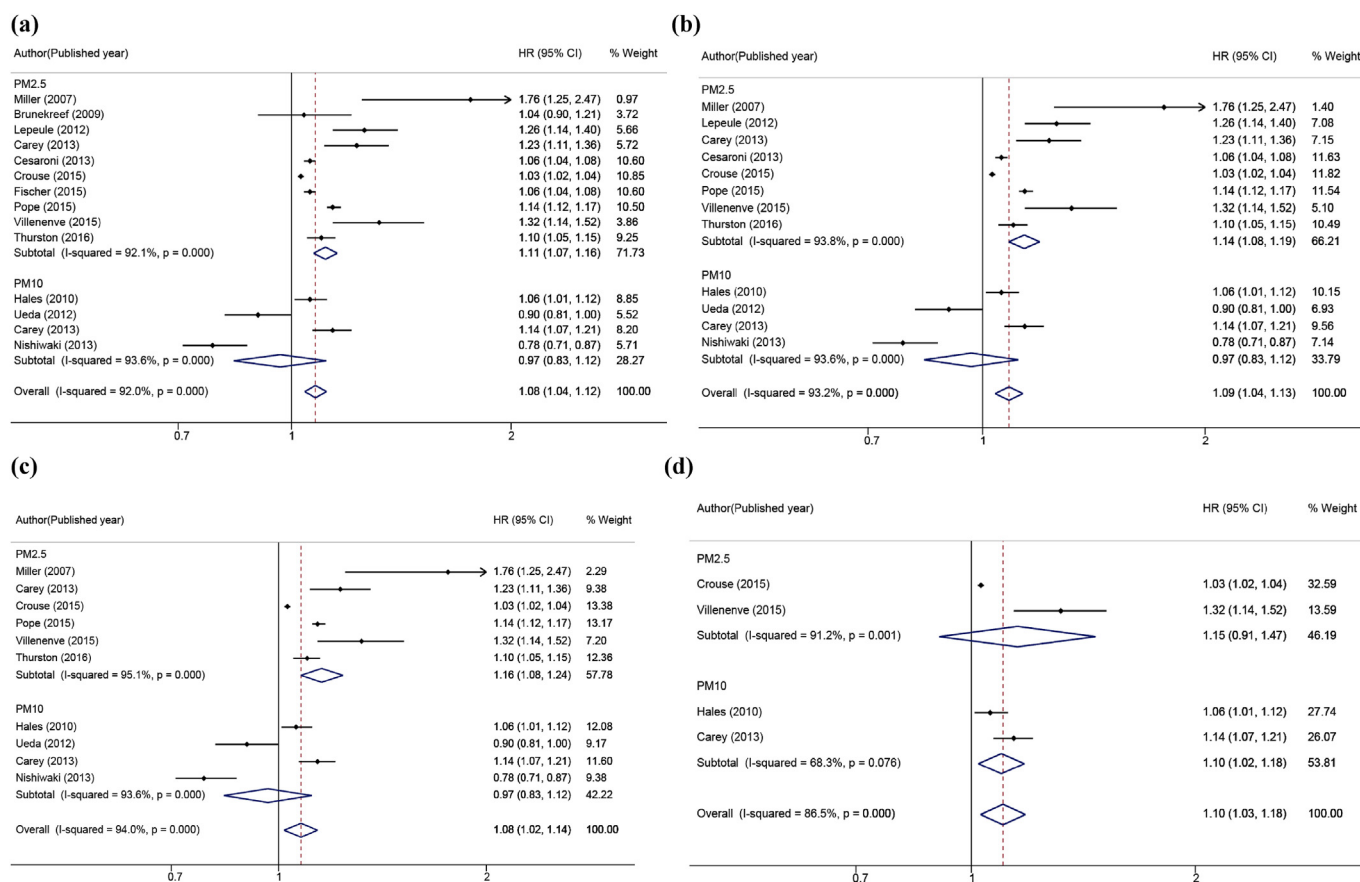


Fig. 3. Meta-analysis: hazard ratios of CVDs mortality for each 10 μ g/m³ increment of PM stratified by annual mean concentration of PM (a) <IT1; (b) <IT2; (c) <IT3; (d) <AQG.

Beelen's review paper as part of sensitivity analysis of our meta-analysis, and the overall results remain unchanged for both PM2.5 (HR = 1.11 VS 1.12) and PM10 (HR = 1.02 VS 1.02) (Supplement 8). There was no evidence for publication bias (Egger's test for asymmetry, $P = 0.192 > 0.1$, Begg's test = 0.662 > 0.1).

4. Discussion

In this study, we systematically reviewed the current state of evidence between ambient PM exposure and CVD mortality in healthy non-occupational population. An increased risk of CVD mortality was observed for long-term PM2.5 exposure but not for PM10 exposure. The RR of CVD mortality for PM exposure did not decrease with the improvement of air quality until the annual PM2.5 concentration was below the AQG.

Previous studies on biological mechanisms revealed that PM2.5 causes CVD events induced by systematic inflammation through reactive oxygen species (ROS) dependent mechanism, as PM2.5 has larger ability to induce ROS than the larger size of PMs (Dick et al., 2003; Martin et al., 1997; Schins et al., 2004; Van Eeden et al., 2012). Furthermore, fine particles can diffuse into circulation and directly damage the endothelium of the circulatory system, which is the major pathogenesis of CVDs (Brook et al., 2003). Pooled effect analyses showed similar risks of CVD mortality in relation to different annual PM2.5 concentrations above AQG of WHO (IT1–3). A lower risk was not observed until air quality had reached the level of AQG. This result conveys an encouraging message that only the strictest standard of ambient PM quality (AQG) might decrease the CVD risk from PM2.5 exposure. Subgroup analysis also showed that longer years of exposure (≥ 11 years) to PM2.5 had an excess effect on

CVD mortality in comparison with the short-term exposure (<11 years), even if their annual PM2.5 concentrations were similar. Hence, the current evidence supports continuous efforts on a long run lowering of the ambient PM exposures at a global level.

Variations in results among studies from different regions might be related to different PM2.5 concentrations, chemical components and genetic backgrounds. The highest HR observed in Asians may be related to its highest PM2.5 concentration (35.3 μ g/m³) (Wong et al., 2015) compared with other included studies (8.9–29 μ g/m³); nevertheless, the effects of PM10 in Asian studies were inconsistent. Results from two Japanese studies (Nishiaki et al., 2013; Ueda et al., 2012) were not supported by studies conducted in other areas (Carey et al., 2013; Hales et al., 2010; LW Zhang et al., 2014). Ueda et al. explained that chemical components of PM10 in Japan have higher proportions of sulfate and element carbon than those in the US and Europe (Ueda et al., 2012). PM is a complex mixture with different emission sources. In developed countries, such as North America and Europe, engine combustion is the major source for PM2.5 (Schauer et al., 1996); however, industrial activity is the major source of emission of PM2.5 in developing countries (Tiwari et al., 2009; Yao et al., 2002). Toxicity of PM2.5 varies if the components of PM2.5 are different. For example, heavy metal particles are more toxic and may cause more serious health effects than other components in PM (Council, 2004). The proportions of heavy metals in PM2.5 differed substantially in the samples taken from different geographic areas (Yang et al., 2005; Ye et al., 2003). Furthermore, genetic background is another plausible reason for such a variation. Previous studies found that the relative risk between genetic scores and CVD events has changed from 1.02 to 1.49 (Sivapalaratnam et al., 2010) and the 2016 European guidelines on

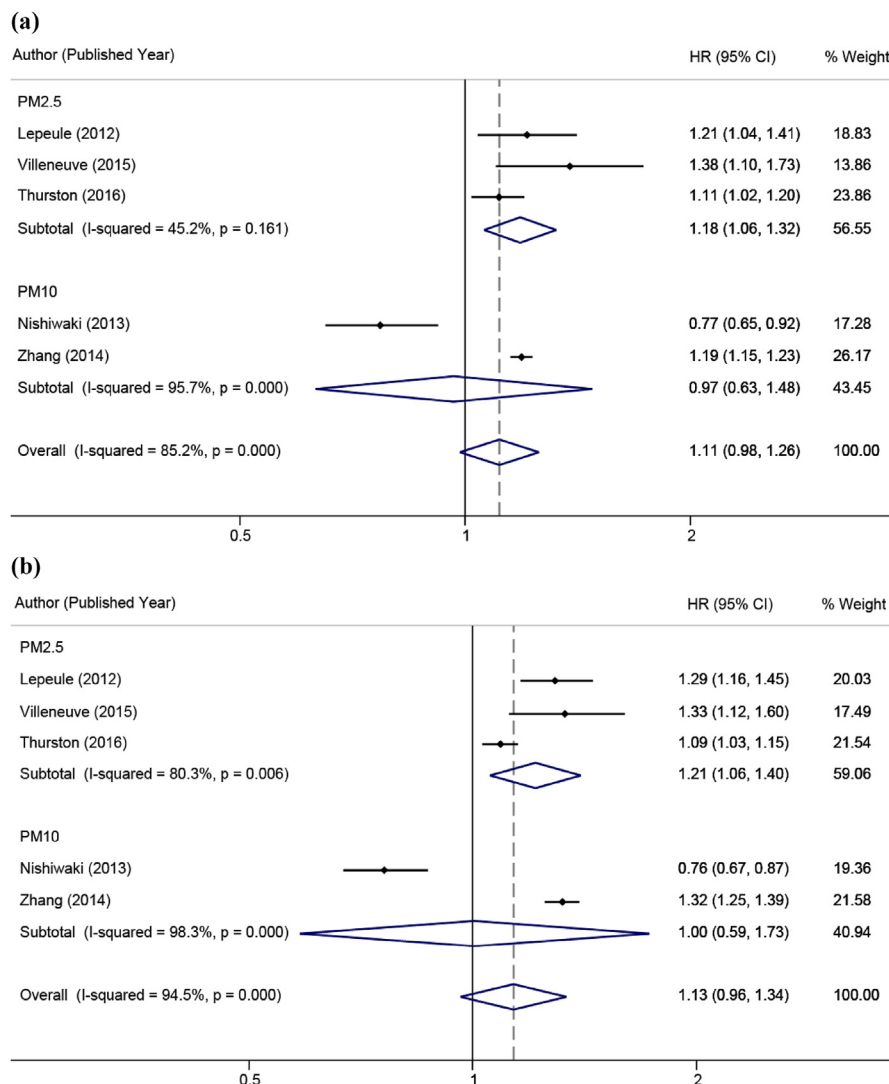


Fig. 4. Meta-analysis: hazard ratios of CVDs mortality for each 10 μ g/m³ increment of PM stratified by smoking status. (a) never smokers; (b) ever smokers (former or current).

cardiovascular disease prevention has listed it as a marker of CVD risk factor (Piepoli et al., 2016).

Importantly, assessment methods of PM exposure might influence the accuracy of results. All studies included in this meta-analysis used surrogate data to rebuild personal PM exposure. The poor method of estimating PM concentration (i.e., category 1) only used one monitoring station data (i.e., assuming same exposure for all residents in the city) without consideration of individual variations regarding location information (Lepeule et al., 2012). Geographic information was considered in estimating PM exposure in many studies, in which researchers tried to use interpolation data from multiple monitoring station or air disposal data (Carey et al., 2013; Cesaroni et al., 2013b; Hales et al., 2010; Miller et al., 2007; Nishiwaki et al., 2013; Ueda et al., 2012; L-w Zhang et al., 2014). With the development of exposure monitoring technology, researchers put atmospheric information into model such as LUR to estimate the concentration of PM and this method could grid geographic cell smaller than the previous 2 methods (Brunekreef et al., 2009; Crouse et al., 2015; Fischer et al., 2015; Pope et al., 2014; Thurston et al., 2016; Villeneuve et al., 2015; Wong et al., 2015). More precise assessment methods of PM_{2.5} exposure, such as personal equipment or portable device, need to be applied in

future environmental epidemiological studies to better quantify the health risk of PM_{2.5}.

This systematic review and meta-analysis on PM and CVD mortality has some merits. We specifically analyzed the long-term effect of ambient PM_{2.5} and PM₁₀ on CVD mortality in subgroups by regions, smoking status, PM assessing methods, exposure length, and exposure levels according to WHO's guidelines that have never been critically assessed and synthesized. However, several limitations of this review should be mentioned. We only focused on studies reporting CVD mortality, therefore, studies only reporting CVD incidences were not included in this meta-analysis. However, as a terminal outcome, mortality is more reliable and less likely to induce misclassifications or under reporting than the incidence. Furthermore, the criteria/classifications of CVD outcomes may vary amongst included studies. The most used criteria for CVD mortality were 390–459 for ICD9 or I00–I99 for ICD10. Several studies used a narrow definition of CVD mortality (Brunekreef et al., 2009; Hales et al., 2010; Lepeule et al., 2012; Nishiwaki et al., 2013), which may introduce a source of heterogeneity when combined all studies together. In general, ambient air quality is much poorer in developing countries than that in the developed countries. However, most of the included studies were

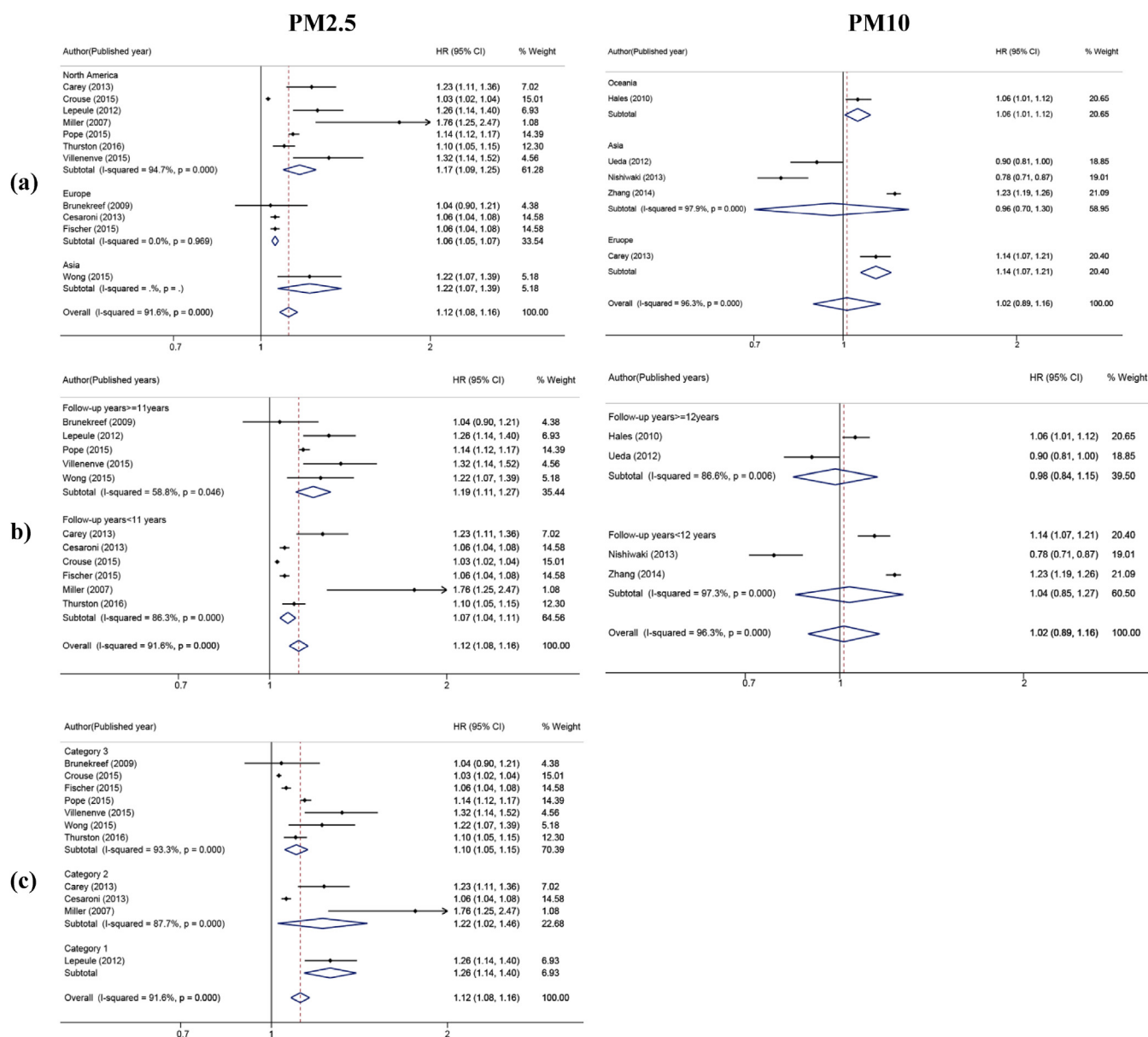


Fig. 5. Meta-analysis: hazard ratios of CVDs mortality for each 10µg/m³ increment of PM stratified by geographic region, follow-up duration, and PM assessment method. Left panels are for PM_{2.5}; Right panels are for PM₁₀; Row (a) geographic regions; Row (b) follow-up years; Row (c) PM assessment methods.

conducted in developed countries. We only found two studies that were conducted in the high PM concentration areas (Hong Kong and China) where the PM concentrations were beyond the IT1 level (Wong et al., 2015; LW Zhang et al., 2014). Evidence showed that 32% of the world population lived in the areas where the air quality was beyond IT-1 (35 µg/m³) exposure level (Brauer et al., 2012). Furthermore, the sources and the components of PM in developing countries differ largely from those of the developed countries. This meta-analysis provides reliable summarized evidence for the developed countries but any attempt to generalize these conclusions to other areas, where the PM concentration is high, should be cautious.

In conclusion, ambient PM_{2.5} exposure level was positively associated with overall mortality of CVD, with a prominent risk in Asian and individuals with long-term exposures. Similar relative risks were reported for studies at lower levels compared to studies with higher exposures. There is no evidence for a threshold, and

this is consistent with many other studies. This study not only emphasizes the importance of exposure assessment on PM at individual level but also calls for more research in Asia and developing countries. More importantly, results from this study echo continuous efforts on a lowering of the ambient PM level to the strictest standard (AQG), particularly for the developing countries.

Conflicts of interest

The authors declare no conflict of interest.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.envpol.2018.07.041>.

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